

LAM KIYO POLY E-CHUCK

Foamtec International
Wilshire Contamination Control Division
cleaning critical surfaces



LAM KIYO POLY E-CHUCK

APPLICATION NOTE
LAM 2300 KIYO
POLY ETCH Chamber PM

OBJECTIVE:

TO EFFECTIVELY PM THE LAM 2300 KIYO POLY ETCH CHAMBER IN A TIMELY MANNER, WHILE IMPROVING TOOL RECOVERY AND PARTICLE PERFORMANCE

Surface Build-up:	PROCESS INDUCED RESIDUE
Tool:	LAM 2300 KIYO POLY ETCH
Tool Parts:	CHAMBER, E-CHUCK AND ASSOCIATED PARTS
Old Method:	1 HOUR USING POLYESTER WIPES AND ALCOHOL (100+ WIPES). Does not remove most residue.
Tool Recovery:	?? **Hours** Conditioning Wafers
New Method:	1 HOUR USING 800D ScrubPAD, DI WATER & IPA
Tool Recovery:	?? **Hours** Conditioning Wafers

Products:

- (1) HT4580D-10-1 Diamond ScrubPAD, 800 Grit
- (1) HT4754 UltraSOLV® Sponge
- (1) FTPEN-1 ScrubWRIGHT™ Pen
- (1) HT4580DW-5 ScrubBelt™ (1 scrubBELT)
- (2) HT5790S-5 MiraWIPES® (10 total wipes)



LAM 2300 KIYO POLY ETCH PM PROCEDURE:

Step 1: Using proper procedures and safety guidelines, shutdown and prepare LAM 2300 Kiyo chamber for wet clean

Step 2: Stage a container of DI water next to the chamber and place a HT4580D 800 grit Diamond ScrubPAD and HT4754 UltraSOLV® Sponge into the container (See Fig 1)

Fig 1: ScrubPAD and UltraSOLV® Sponge in container of DI



NOTE: MAY SUBSTITUE A SMALL AMOUNT OF DI WATER or 90/10 TO MOISTEN UltraSOLV® SPONGE AND ScrubPAD AS NECESSARY (See Fig 2 & 3)



Fig 2: UltraSOLV® Sponge, ScrubPAD and Small Amount of DI Water

Fig 3: Dampening UltraSOLV® Sponge with squirt bottle



Step 3: Wipe down and place protective cover over E-Chuck top surface to protect it (See Fig 4)

Fig 4: Protective cover on E-Chuck



LAM 2300 KIYO POLY ETCH PM PROCEDURE (CONT):

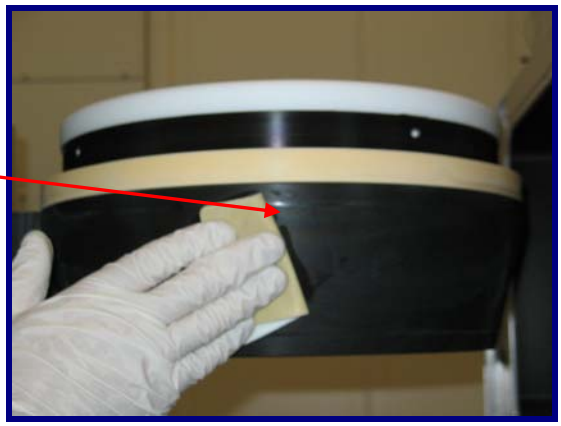
Step 4: Take **lightly dampened** UltraSOLV® Sponge and begin to wipe E-Chuck sides (See Fig 5)

Fig 5 : Lightly dampened UltraSOLV® sponge wiping side of E-Chuck



Step 5: Take **lightly dampened** 800D Grit Diamond ScrubPAD and proceed to scrub off deposition from POLY ETCH E-Chuck sides. Important to keep area a little moist with DI Water and to scrub a small area about 4" x 4" at a time (See Fig 6)

Fig 6: Lightly Dampened 800D Grit ScrubPAD Scrubbing E-Chuck side, keeping slightly moist with DI water



NOTE: Only scrub from the ceramic ring down. DO NOT scrub top portion of E-Chuck.

Step 6: Before deposition has a chance to dry back onto the E-Chuck, take lightly dampened UltraSOLV® Sponge and wipe deposition off of E-Chuck side (See Fig 7)



Fig 7: Lightly Dampened UltraSOLV® Sponge wiping E-Chuck side

LAM 2300 KIYO POLY ETCH PM PROCEDURE (CONT):

Step 7: As Diamond ScrubPAD appears to load up with process residue, pull ScrubPAD across **damp** UltraSOLV® Sponge. This will help keep ScrubPAD effective at removing process residue from chamber liner (See Fig 8, 9 & 10)

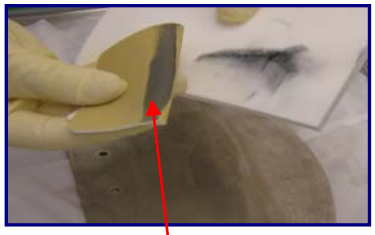


Fig 8: Diamond ScrubPAD loaded with process residue

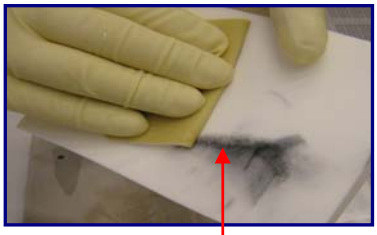


Fig 9: Diamond ScrubPAD loaded with process residue

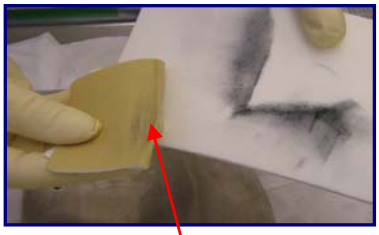


Fig 10: Unloaded Diamond ScrubPAD

Step 8: Continue to Rinse UltraSOLV® Sponge in Container of DI Water as necessary to keep UltraSOLV® Sponge slightly moist and free of process residues (See Fig 11)

Fig 11: Rinsing UltraSOLV® Sponge with DI water as necessary



LAM 2300 KIYO POLY ETCH PM PROCEDURE (CONT):

Step 9: Continue to repeat this **SCRUB – WIPE – RINSE** procedure outlined in Steps 4 thru 8 for the remainder of the LAM 2300 KIYO POLY Chamber (See Fig 11, 12 & 13)



Fig 11: Clean stains off of ceramic ring



Fig 12: Clean process residue around chamber entrance

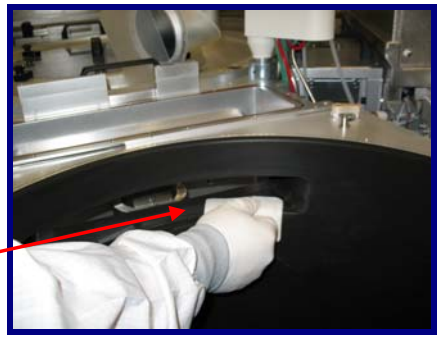


Fig 12: Clean process residue around entire Chamber





Fig 13: Cleaned chamber

LAM 2300 KIYO POLY ETCH PM PROCEDURE (CONT):

IMPORTANT NOTE

THE USE OF HT5790S MiraWIPES® DURING THE FINAL WIPE PORTION OF THE PROCEDURE IS A CRITICAL STEP TO EFFECTIVELY REMOVE PARTICLE DEFECTS AND MOISTURE FROM THE SOURCE CHAMBER

NOTE: Below is an example of the particles left behind in a Process Chamber after the final wipe portion of the PM was performed with the **Current Fab Wiper.** (See Fig 14a & 14b)

Fig 14a: Current Fab Wiper after completely wiping chamber



Fig 14b: Particles picked up using HT5790S MiraWIPES® after completely wiping with Current Fab Wiper.

MiraWIPES are the KEY STEP for DEFECT REDUCTION and IMPROVE TOOL RECOVERY

Step 10: After scrubbing the Chamber and E-Chuck change to a new pair of gloves prior to final wipe down

Step 11: Fold a MiraWIPE® into quarters and soak with IPA. Proceed to wipe down entire Process Chamber and E-Chuck (see Fig 15)



Fig 15: Wiping E-Chuck with MiraWIPE®s.

TOOL RECOVERY:

Step 1: Follow proper Tool Recovery guidelines as outlined by LAM Research Corporation

